

Supplementary raw data on cell chaining in an attenuated *Listeria monocytogenes*

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Abstract

The dataset "Supplementary raw data on cell chaining in an attenuated *Listeria monocytogenes*" provides the raw images and data for the manuscript "On the role of cell chaining in the attenuation of a *Listeria monocytogenes* divIVA mutant" (<https://doi.org/10.64898/2026.01.05.697648>).

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Information on the dataset and context of origin

The dataset “Supplementary raw data on cell chaining in an attenuated *Listeria monocytogenes*” provides the raw images and data, which form the basis for the figures of the manuscript “On the role of cell chaining in the attenuation of a *Listeria monocytogenes* divIVA mutant”. In-depth information on the background of the work, on methodology and on the analysis of the results can be found in

Wamp, S., Holland, G., Rismondo, J., and Halbedel, S. (2026). On the role of cell chaining in the attenuation of a *Listeria monocytogenes* divIVA mutant. bioRxiv.
<https://doi.org/10.64898/2026.01.05.697648>

Administrative and organizational information

The publicaton “Supplementary raw data on cell chaining in an attenuated *Listeria monocytogenes*”, is a joined work of the Consultant Laboratory for Listeria in [Unit 11| Enteropathogenic Bacteria and Legionella](#) of the Robert Koch Institute, the Unit [ZBS 4| Advanced Light and Electron Microscopy](#) and the Institute for [Medical Microbiology and Hospital Hygiene](#) at the Otto von Guericke University. Questions regarding the content of the data can be addressed directly to the corresponding author Sven Halbedel (halbedels@rki.de).

The publication of the data as well as the quality management of the (meta-)data is done by the unit [MF 4 | Domain Specific Data and Research Data Management](#). Questions regarding data management and the publication infrastructure can be directed to the Open Data Team of the unit MF4 at OpenData@rki.de.

Overview over the *L. monocytogenes* strains used for this manuscript

The following strains were used for this manuscript. Detailed methodology on their generation can be found in the methods section of the manuscript (<https://doi.org/10.64898/2026.01.05.697648>).

Strain name	Genetic background	Description
EGD-e	wt	Wildtype strain <i>L. monocytogenes</i> , used as control
LMSW49	<i>imreB</i>	MreB depletion strain, MreB expression can be induced upon IPTG-treatment
LMJR183	<i>iezrA</i>	EzrA depletion strain, EzrA expression can be induced upon IPTG-treatment
LMS2	Δ <i>divIVA</i>	divIVA deletion strain
LMSW207	Δ <i>divIVA imreB</i>	divIVA deletion and MreB depletion strain, MreB expression can be induced upon IPTG-treatment

Strain name	Genetic background	Description
LMSW208	$\Delta divIVA$ <i>iezrA</i>	<i>divIVA</i> deletion and <i>EzrA</i> depletion strain, <i>EzrA</i> expression can be induced upon IPTG-treatment
BUG2214	$\Delta prfA$	<i>prfA</i> deletion strain
LMS30	<i>idivIVA</i>	<i>divIVA</i> depletion strain, <i>divIVA</i> expression can be induced upon IPTG-treatment
LMJD20	wt	DsRed producing <i>L. monocytogenes</i> strain, used as control
LMSF2	$\Delta actA$	DsRed producing, <i>actA</i> deletion strain
LMSW205	$\Delta divIVA$	DsRed producing, <i>divIVA</i> deletion strain
LMSW202	<i>iezrA</i>	DsRed producing, <i>EzrA</i> depletion strain, <i>EzrA</i> expression can be induced upon IPTG-treatment
LMSW203	<i>imreB</i>	DsRed producing, <i>MreB</i> depletion strain, <i>MreB</i> expression can be induced upon IPTG-treatment
BUG3057	<i>prfA</i> *	<i>L. monocytogenes</i> strain that encodes a <i>PrfA</i> variant that is constitutively active causing constitutive overexpression of <i>PrfA</i> -dependent virulence genes
LMSW251	$\Delta divIVA$ <i>prfA</i> *	<i>divIVA</i> deletion strain that encodes a <i>PrfA</i> variant that is constitutively active causing constitutive overexpression of <i>PrfA</i> -dependent virulence genes
LMSW222	$\Delta divIVA$ <i>secA2</i> M95I	<i>divIVA</i> deletion strain with <i>secA2</i> mutation (M95I)
LMSW223	$\Delta divIVA$ <i>secA2</i> T123A	<i>divIVA</i> deletion strain with <i>secA2</i> mutation (T123A)

Content and structure of the raw supplemented raw images and data

The raw data and relevant calculations that form the basis for the manuscript's figures are provided as .xlsx files.

The raw microscopy images, as well as uncropped images from a western blot, a gel electrophoresis and culture plates are provided in the "Images" subfolder as .tif files.

> [Images/](#)

The naming and content of the image and data files are explained in the tables below per figure.

Figure 1

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Figure 1B	Uncropped Western blot showing MreB and DivIVA expression levels in the strains EGD-e (wt) and LMSW49 (<i>imreB</i>) $\pm$ IPTG	Images/Figure_1B_uncropped_blot_images.tif
Figure 1C	Fluorescence microscopy image of EGD-e (wt, Nile red)	Images/Figure_1C_wt_raw_image.tif
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Figure 1C	Fluorescence microscopy image of LMSW49 (<i>imreB</i>) without IPTG (Nile red)	Images/Figure_1C_imreB-IPTG_raw_image.tif

Figure 2

Figure	Description	File
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Figure	Description	File
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Figure	Description	File
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Figure 4

Figure	Description	File
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Figure 4	Scanning electron microscopy image of LMSW49 (<i>imrB</i>) without IPTG	TODO: rename image? Images/Figure_4_wt_imrB_image.tif

Figure	Description	File
Figure 4	Scanning electron microscopy image of LMJR183 (<i>iezrA</i>) without IPTG	TODO: rename image? Images/Figure_4_wt_ezrA_image.tif

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Figure	Description	File
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Figure 6

Figure	Description	File
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Figure	Description	File
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Figure 7

Figure	Description	File
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Figure 8

Figure	Description	File
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Figure	Description	File
	<i>secA2 M95I</i> , Nile red)	
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Supplementary Figure S1

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Figure S1	Scanning electron microscopy image of LMS2 ($\Delta divIVA$)	Images/Figure_S1_divIVA_raw_image.tif
Figure S1	Scanning electron microscopy image of LMJR183 (<i>iezrA</i>)	Images/Figure_S1_ezrA_raw_image.tif

Figure	Description	File
Figure S1	Scanning electron microscopy image of LMSW49 (<i>imreB</i>)	Images/Figure_S1_mreB_raw_image.tif

Supplementary Figure S2

Figure	Description	File
Figure S2	Growth curves of the strains LMSW207 ($\Delta divIVA$ <i>imreB</i>) and LMSW208 ($\Delta divIVA$ <i>iezrA</i>)	Figure_S2_raw_data.xlsx

Supplementary Figure S3

Figure	Description	File
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Figure	Description	File
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Supplementary Figure S4

Figure	Description	File
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Figure	Description	File
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Metadata

To increase findability, the provided data are described with metadata. The Metadata are distributed to the relevant platforms via GitHub Actions. There is a specific metadata file for each platform; these are stored in the metadata folder:

[Metadata/](#)

Versioning and DOI assignment are performed via [Zenodo.org](#). The metadata prepared for import into Zenodo are stored in the [zenodo.json](#). Documentation of the individual metadata variables can be found at <https://developers.zenodo.org/representation>.

[Metadata/zenodo.json](#)

The zenodo.json includes the publication date and the date of the data status in the following format (example):

```
"publication_date": "2024-06-19",
"dates": [
  {
    "start": "2023-09-11T15:00:21+02:00",
    "end": "2023-09-11T15:00:21+02:00",
    "type": "Created",
    "description": "Date when the published data was created"
  }
],
```

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- <https://github.com/robert-koch-institut>
- <https://gitlab.opencode.de/robert-koch-institut>

- <https://edoc.rki.de/>

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